ADVANCED DATA ANALYSIS · MODULE III

Causal Inference and Machine Learning

Why a good predictor is not a causal estimate

Week 10

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THE STARTING POINT

Two kinds of questions

Prediction

“Who will be hospitalised next year?”

We want a good guess about new cases. We can check the guess: hide part of the data, predict it, and compare. Any variable that improves the guess is welcome.

Causation

“Do hospitals make people healthier?”

We want to know what would change if we intervened. We can never check this directly: no one is seen both with and without the hospital. A variable that improves the guess can even mislead us.

Weeks 7–9 were about the left card. Today is about the right one.

PART 1 · THE PROBLEM

Do hospitals make people healthier?

A large US health survey asked two things: “Did you stay in hospital last year?” and “How is your health?” (1 = excellent … 5 = poor).

Group	People	Average health
Stayed in hospital	7,774	2.79
Did not	90,049	2.07

THE GAP

0.71

Hospital patients report worse health. The gap is far too large to be chance.

> **So do hospitals make people sick?**
>
> No. People go to hospital *because* they are already sick. The two groups were different before anyone was treated.

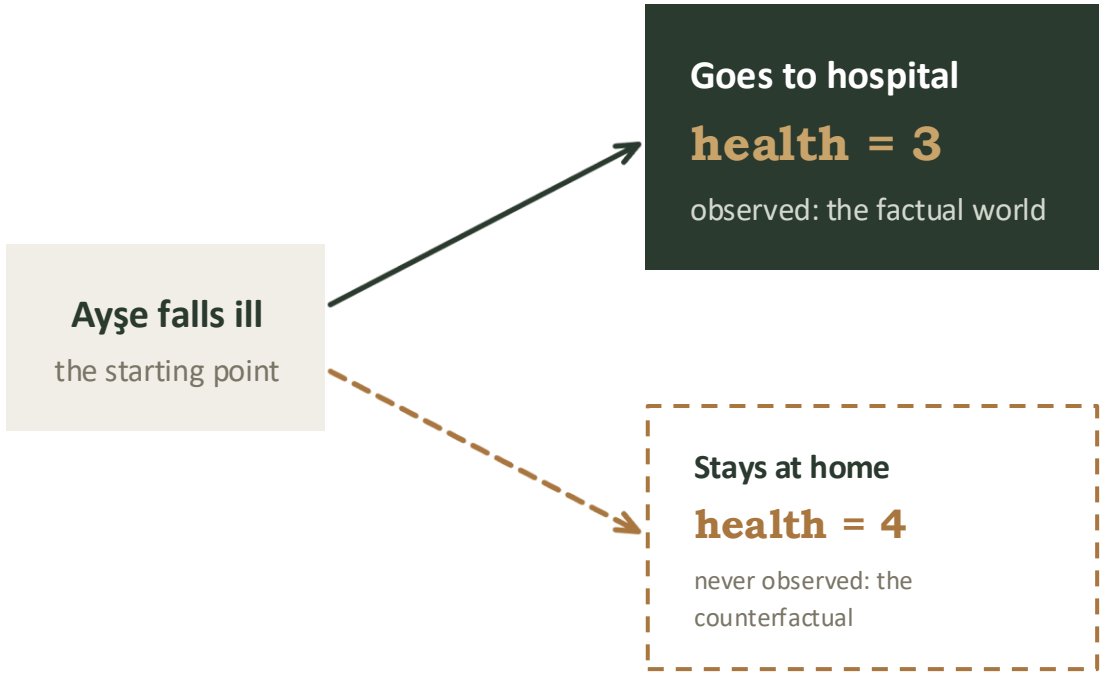
Data: US National Health Interview Survey, as reported in this week's reading (Angrist & Pischke, ch. 2).

PART 1 · THE PROBLEM

Causality means a counterfactual

A causal claim always compares two worlds: the one that happened and the one that did not.

Factual	What actually happened: Ayşe went to hospital.
Counterfactual	What would have happened to the same person, at the same time, had she not gone.
Causal effect	The difference between the two worlds: 3 − 4 = −1, one point healthier.



We never see both worlds for the same person. Every causal method is a way of estimating the missing one.

PART 1 · THE PROBLEM

Potential outcomes: one for each world

Every person has two **potential outcomes**, written down before we know which world happens:

Y_1 health **with** treatment (if hospitalised)

Y_0 health **without** treatment

D which world actually happened (1 = hospital)

Observed or counterfactual?

If D = 1, we observe Y_1 and Y_0 is the counterfactual. If D = 0, it is the other way round. The individual effect is $Y_1 - Y_0$.

What the data look like:

Person	D	Y_1	Y_0
Ayşe	1	3	?
Mehmet	1	2	?
Zeynep	0	?	1
Can	0	?	2

Every **?** is a counterfactual. Causal inference is the art of filling them in credibly.

A made-up example with four people. Health: 1 = excellent … 5 = poor, so lower is better.

PART 1 · THE PROBLEM

Why the simple comparison misleads

Suppose we could see the counterfactuals (in copper):

Person	D	Y_1	Y_0
Ayşe	1	3	4
Mehmet	1	2	3
Zeynep	0	1	1
Can	0	2	2

Ayşe and Mehmet were sicker to begin with (Y_0 = 4 and 3). The hospital improved each of them by one point.

What we see
patients' average minus others': 2.5 − 1.5
+1.0

=

The true effect on patients
$Y_1 - Y_0$ for those who went
−1.0

+

Selection bias
how different the groups were anyway: Y_0 of 3.5 vs 1.5
+2.0

With observational data, people choose their treatment, and those who choose it are different to begin with.

PART 1 · THE EXPERIMENTAL IDEAL

Why experiments solve the problem

STEP 1

A coin decides

Who goes to hospital depends on chance, not on how sick people are.

STEP 2

The groups start out alike

On average they have the same Y_0: same age, illness, income, and even what we cannot measure.

STEP 3

The gap is the effect

The control group shows the treated group's counterfactual. Selection bias is zero.

The same idea as a regression: $Y = \alpha + \rho \cdot D + \text{error}$

When D is random, ρ is the causal effect. Adding control variables should barely change ρ; if it does, something is wrong with the randomisation.

No experiment possible? Sometimes the world runs one for us. Next: natural experiments.

PART 1 · NATURAL EXPERIMENTS

When chance happens anyway

Quasi-experimental designs look for variation in treatment that is **as good as random**, even though no researcher assigned it. Each design finds its "coin flip" somewhere else:

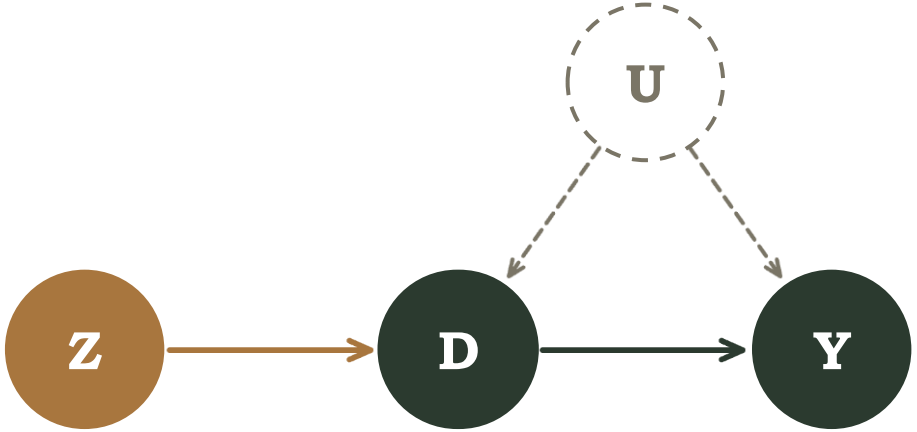
Instrumental variable	Something that nudges people into treatment, but affects the outcome only through it.
Difference-in-differences	A change that hits one group and not another, at a known time.
Event study	The same comparison, traced period by period before and after the event.
Regression discontinuity	A rule with a cutoff: people just above and just below it are alike.
Synthetic control	One treated unit, compared with a weighted blend of similar untreated units.

The question stays the same in every design: where does the counterfactual come from?

PART 1 · NATURAL EXPERIMENTS

Instrumental variables

Does college raise earnings? Ability and motivation (U) drive both, and we cannot measure them.



Z: a college in your town · D: going to college · Y: earnings · U: ability, motivation (unobserved). No arrow from Z to Y.

The idea: some people go to college *only because* one happens to be nearby. That part of college-going is unrelated to ability.

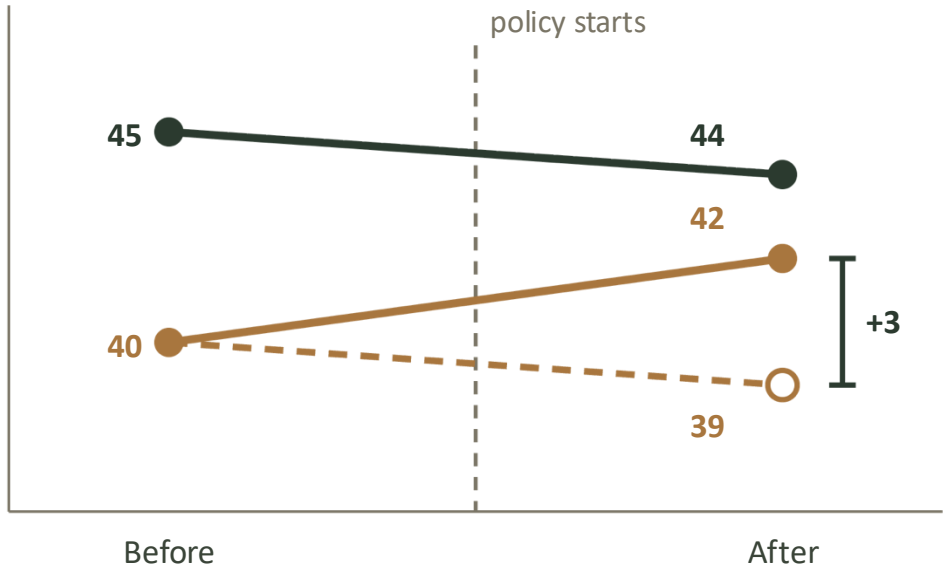
Relevance	Z really changes D: a nearby college makes attending more likely.
Exclusion	Z affects Y only through D. Debatable here: college towns may also have better jobs.
As if random	Z is unrelated to U, perhaps after controlling for region.

Effect = (Z's effect on Y) ÷ (Z's effect on D)

e.g. earnings +2%, attendance +10 points → 2 ÷ 0.10 = 20% per college degree, for people whom Z moved.

Illustrative numbers. The estimate applies to "compliers": people whose choice was changed by the instrument.

Difference-in-differences



City A raised the minimum wage · **City B** did not · dashed: A's counterfactual, moving like B

Does a higher minimum wage cost young people jobs? Compare how youth employment (%) changed in A with how it changed in B.

The estimate

$(42 - 40) - (44 - 45) = 2 - (-1) = 3$

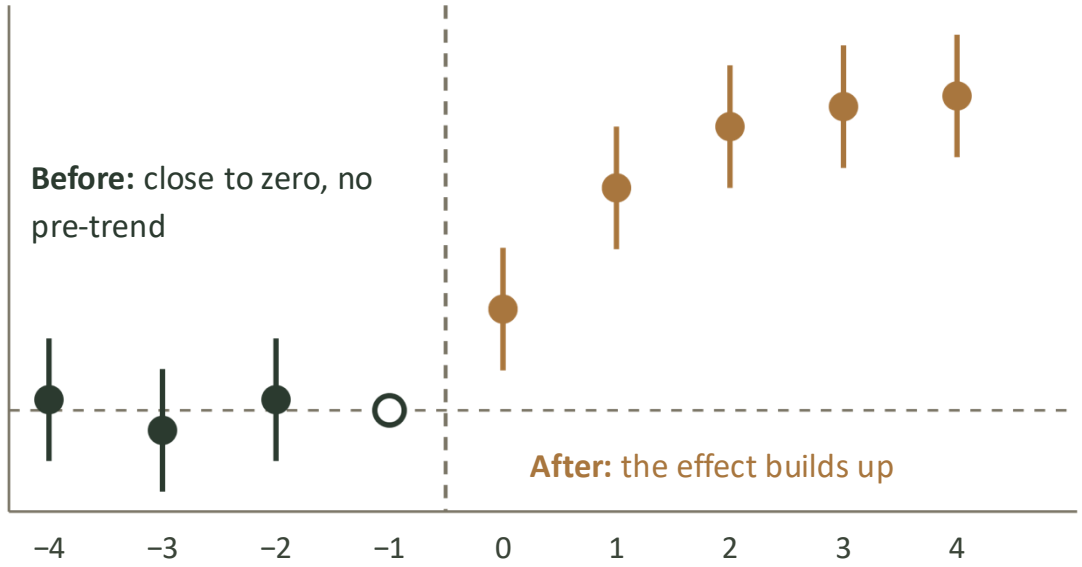
A's change minus B's change: the first difference removes fixed gaps between the cities, the second removes shocks that hit both.

Key assumption: parallel trends

Without the policy, A would have moved like B. B's trend is A's counterfactual.

Illustrative numbers. In a regression: $Y = \alpha + \beta \cdot \text{treated} + \gamma \cdot \text{after} + \delta \cdot (\text{treated} \times \text{after})$; δ is the effect.

Event studies with panel data



Years relative to the event (−1 is the reference year); bars show 95% intervals

Example: factories close in several towns, in different years. How does local unemployment evolve?

Instead of one before/after gap, estimate the treated–control gap for **every year**, relative to the year before the event.

Before the event

Estimates near zero support parallel trends. A trend here is a warning.

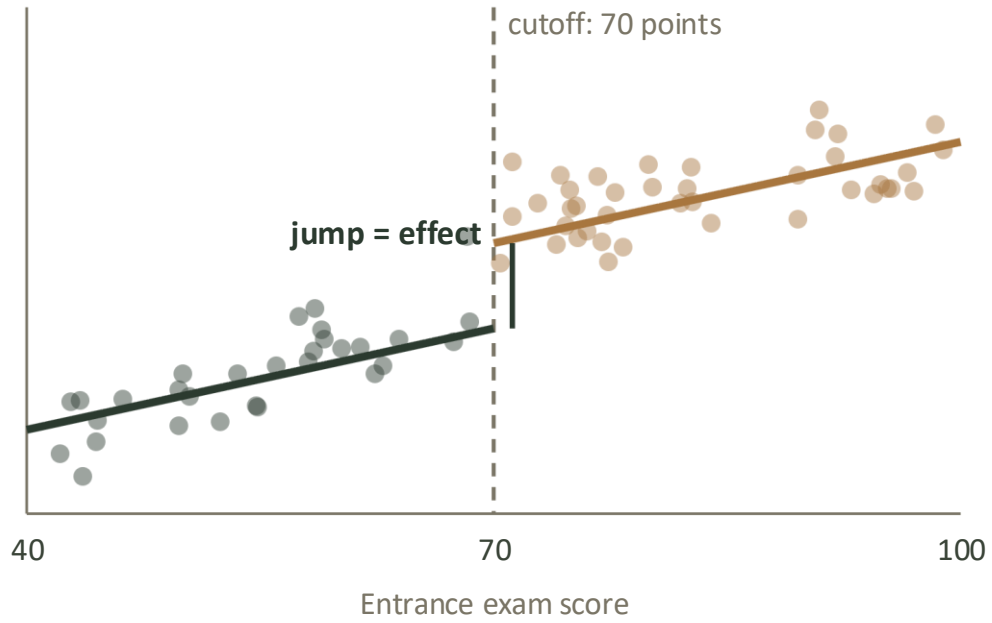
After the event

Shows whether the effect is immediate, grows or fades.

Illustrative estimates. Panel data: the same units observed every year; units may be treated at different times.

PART 1 · NATURAL EXPERIMENTS

Regression discontinuity



Vertical axis: share completing university (%) · **below 70:** no scholarship · **70 and above:** scholarship

Does a scholarship help students finish university? The rule: everyone scoring 70 or more gets one.

Students with 69 and 71 points are almost identical, except that one got the scholarship. Just below the cutoff is the counterfactual for just above.

The estimate

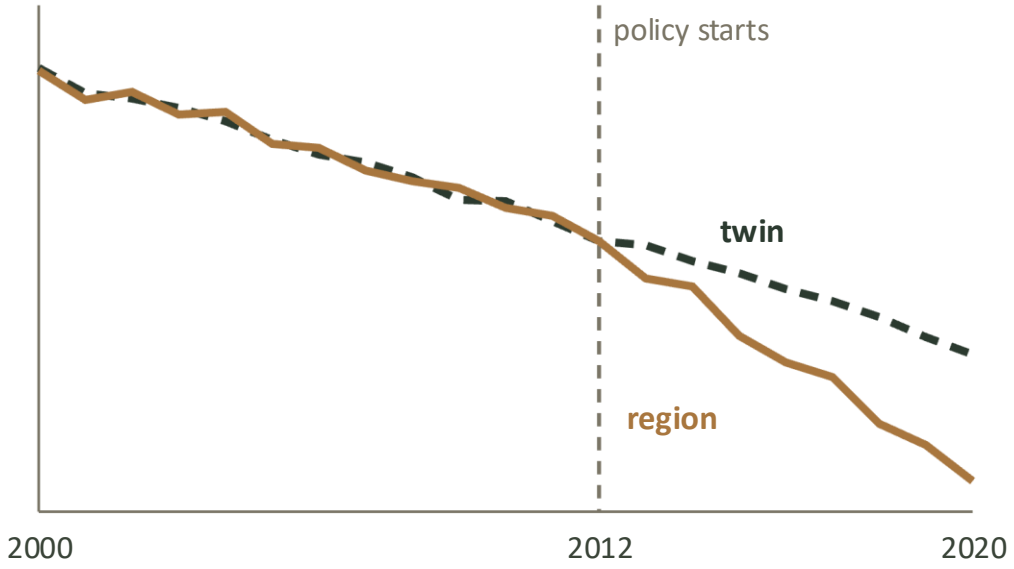
Fit a line on each side; the jump at 70 (here 42% → 52%, about 10 points) is the effect.

Assumptions and limits

No one can precisely push their score over 70, and nothing else changes there. The effect holds near the cutoff.

Simulated data. Check for bunching just above the cutoff: it signals that people manipulated the score.

Synthetic control



Solid: the region with the policy · **dashed:** its synthetic twin, e.g. 0.5 × B + 0.3 × C + 0.2 × D

Did a regional anti-smoking law cut cigarette sales? Only one region is treated, and no other region looks exactly like it.

1 · Build a twin

Weight untreated regions so that their blend tracks the treated region before the policy.

2 · Read the gap

After 2012 the twin shows the counterfactual; the gap is the effect.

3 · Check it

Repeat for untreated regions as placebos: their gaps should be small.

Simulated data. Choosing the weights is an optimisation problem, a first bridge to machine learning.

PART 1 · SUMMARY

Where the counterfactual comes from

Design	Counterfactual from	Key assumption	Effect for whom
Experiment	the randomised control group	randomisation worked	study participants
Instrumental variable	people not nudged by Z	Z affects Y only via D	those Z moved
Diff-in-differences	the untreated group's trend	parallel trends	the treated group
Event study	the same, year by year	parallel trends, no anticipation	the treated, over time
Discontinuity	units just below the cutoff	no sorting at the cutoff	units near the cutoff
Synthetic control	a weighted twin	the twin fits before	the one treated unit
Adjusting for X	similar people who were untreated	no hidden confounders	depends on overlap

Often no natural experiment is available. Then we adjust for X: the last row, and the subject of Part 2.

PART 2 · MACHINE LEARNING

A counterfactual is a prediction

Look back at Part 1. Every design predicts the same missing thing: **the outcome the treated would have had without the treatment.**

DESIGN	PREDICTS THE MISSING WORLD FROM
Experiment	the control group
Difference-in-differences	the untreated group's trend
Synthetic control	a weighted blend of other regions
Adjusting for X	similar people who were not treated

So machine learning can help

ML is built to predict. Filling in the missing world is a prediction task, and a hard one when many variables matter.

But there is a twist

In week 8 we checked predictions on a test set. Here the truth is never observed, so no test set can tell us whether the counterfactual is right. The design and its assumptions must do that job.

Machine learning fills in the missing world. The research design tells us whether the filling is believable.

PART 2 · WHAT CHANGES FROM WEEK 8

Same tools, different job

	Prediction (weeks 7–9)	Causal inference (today)
Goal	predict Y for new cases	predict the missing counterfactual, then compare
The model is	the final product	a helper step inside the estimate
We check it with	held-out test data	no test data exists: design and assumptions
Good variables	anything that predicts Y	confounders only; nothing caused by the treatment
We report	accuracy, F1, RMSE	the effect, its uncertainty, the assumptions

Beware of bad controls

Length of hospital stay predicts health very well, but the hospital itself causes it. If we control for it, we erase part of the very effect we want to measure.

PART 2 · A MAP

Four jobs for machine learning

1 Adjust for many confounders

With no natural experiment, predict the outcome and the treatment from dozens of controls, curves and interactions included.

Tools: lasso, random forests, double ML

2 Sharpen natural experiments

IV, DiD and synthetic control contain prediction steps too: the first stage of IV, the synthetic twin.

Tools: lasso, elastic net, matrix completion

3 Find who is affected

Predict how large the effect is for each kind of person, instead of one average for everyone.

Tools: causal trees, causal forests

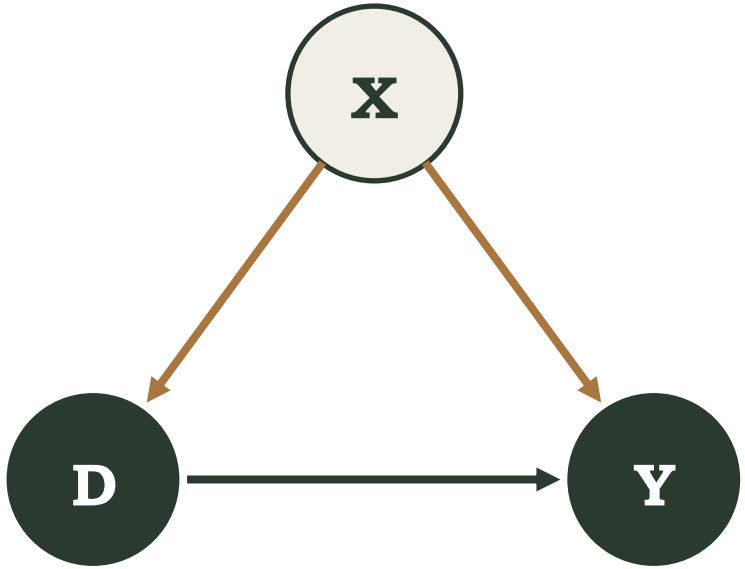
4 Measure from text and images

Turn articles, posts and speeches into variables that serve as treatment, outcome or control.

Tools: classifiers, embeddings, LLMs (Module IV)

In every job, ML does a prediction task. Whether the result is causal is decided by the design, not by the algorithm.

When we cannot run an experiment



X: age, earlier illness, income · D: going to hospital · Y: health. X pushes both D and Y.

The idea: compare people who are alike in X. Among people with the same age, illness history and income, whether someone went to hospital may be close to chance. Then the untreated show us the counterfactual of the treated.

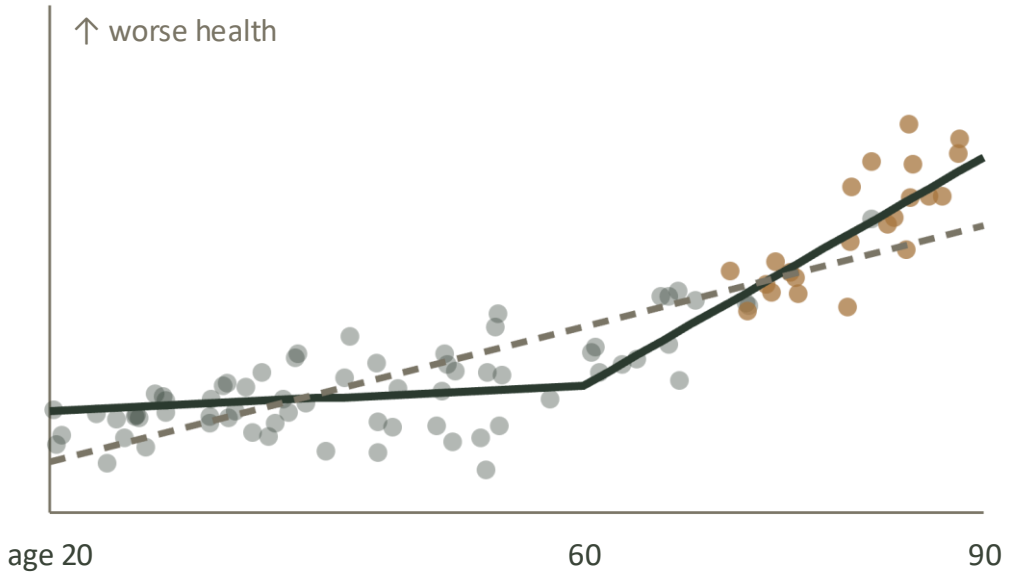
No hidden confounders	Everything that pushes both hospital use and health is in X. We cannot test this; we have to argue for it.
Overlap	For every kind of person, some went to hospital and some did not.

Where ML comes in

X may hold fifty variables with curved effects and interactions. Writing that model by hand is hopeless; ML can learn it from the data.

The second reading writes W for the treatment; we keep D throughout.

When a straight line is not enough



● hospital patients ● others solid line: true age curve dashed: straight-line control

Health worsens slowly until 60, then fast. Older people also go to hospital more often.

A straight line misses the bend
Controlling for age with a straight line under-corrects exactly where the oldest people are.

The leftover is blamed on the hospital
Most patients are old, so the missed part of the age effect looks like a hospital effect.

ML learns the shape
Trees and forests find the bend, and interactions such as age × chronic illness, without us writing them down.

Simulated data. With one control we could fix this by hand; with fifty, we cannot.

Lasso needs two looks

Lasso (week 8) is handy when there are many controls: it keeps the variables that help predict and drops the rest. In causal work the question is: *predict what?*

The trap

Health insurance barely predicts health, so a lasso that predicts health drops it. But insurance strongly decides who goes to hospital. Leaving it out mixes the insurance effect into the hospital effect: **omitted-variable bias**.

The fix: double selection

1. Lasso 1: which controls predict **health (Y)**?
2. Lasso 2: which controls predict **going to hospital (D)**?
3. Ordinary regression of Y on D, controlling for every variable picked in step 1 **or** step 2.

We keep every confounder that matters for either side, so none slips through.

Rule of thumb: a variable that predicts the treatment matters, even if it barely predicts the outcome.

Double machine learning in three steps

1

Predict the outcome

ML predicts health from X. Keep the leftover (residual): the part of health X cannot explain.

2

Predict the treatment

ML predicts hospital use from X (the propensity score). Keep the leftover: the part X cannot explain.

3

Compare the leftovers

Regress leftover health on leftover hospital use. The slope is the hospital effect.

Why it works

The leftovers no longer contain X, so X cannot fake a link between hospital and health.

Why "double"

Two ML models, one for each side. A small mistake in either barely moves the final estimate.

CROSS-FITTING, FIVE FOLDS



Each person's leftover comes from a model that never saw that person, like a test set in week 8.

Lasso in the first stage of IV

Two-stage least squares (2SLS) has two stages. **Stage 1 is pure prediction:** how well do the instruments Z predict the treatment D? **Stage 2 is the causal part.**

Many candidate instruments

$Z_1, Z_2, \ldots Z_{50}$

e.g. dozens of judge traits

→

Stage 1 · prediction

Lasso keeps the few Z that really predict D and produces a **predicted D**.

→

Stage 2 · causal

Use the predicted D in place of D to estimate its effect on Y.

→

Effect on Y

for people whom Z moved

Why lasso?

With many instruments, an ordinary first stage overfits: the predicted D copies D, confounded part included, and the estimate drifts back toward the biased simple regression. Lasso keeps only the strong instruments.

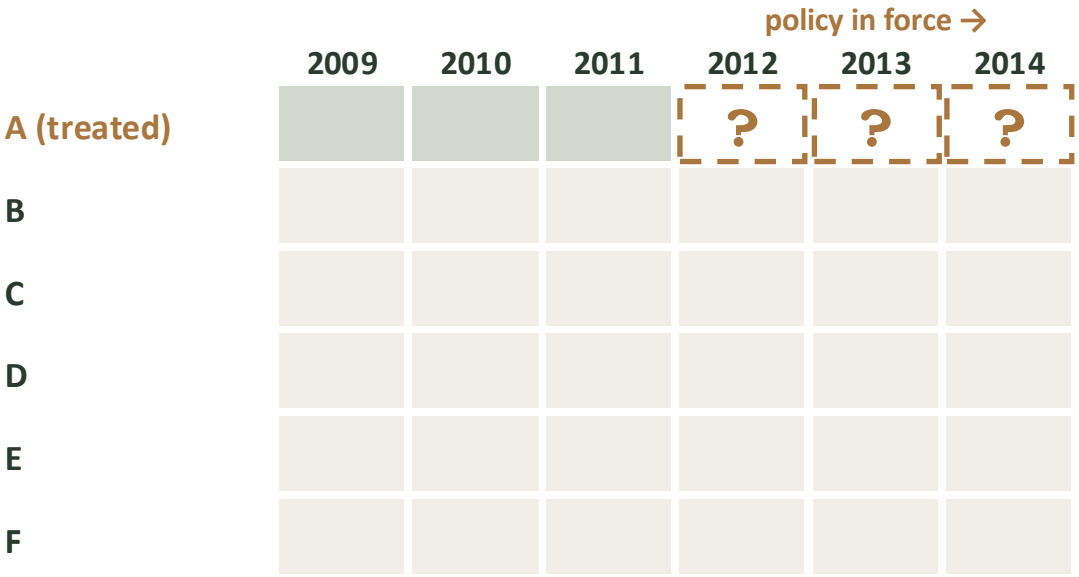
What lasso cannot do

It only checks relevance: does Z predict D? Whether each Z affects Y only through D (exclusion) is still an argument, not a computation.

Example: cases are assigned to judges at random; judge traits (Z) predict how strict a judge is, and strictness decides detention (D).

Better synthetic twins with ML

The counterfactual is a set of missing cells:



Filled cells are observed. The **?** cells are A without the policy: we fill them in, much like predicting the ratings you would give films you have not watched.

Classic synthetic control

Weights for donor regions (positive, summing to one) chosen to match A before the policy.

The same, as prediction

Predict A from all donor regions with lasso, ridge or elastic net. Many donors are fine; the penalty stops overfitting.

Matrix completion

Fill in every missing cell at once, using patterns shared across regions and years.

DiD and event studies

ML adjusts for covariates that make parallel trends more believable (doubly robust DiD).

A test set for the counterfactual: hide the last pre-policy years, predict them, and check the fit before trusting the twin.

PART 2 · JOB 3 · FINDING WHO IS AFFECTED

The effect is not the same for everyone

So far we estimated one number. But a hospital stay may help an 80-year-old a lot and a 30-year-old hardly at all. Three averages answer three questions:

Average effect (ATE)	for everyone
Effect on the treated (ATT)	for those who actually went
Conditional effect (CATE)	for a group, e.g. over-65s

How machine learning helps

Causal tree
Splits people into groups where the effect differs most, then estimates each group's effect on data it did not use for splitting.

Causal forest
Averages many such trees and gives an estimated effect for every person.

Targeting
Uses these estimates to decide who should get a programme when resources are limited.

Example from the reading: a college degree reduces low-wage work most for people from disadvantaged backgrounds.

PART 2 · JOB 4 · MEASURING FROM TEXT

Text and images as variables

Many social-science variables live in text: how a protest was framed, how hostile a reply is, what a speech is about. ML turns them into numbers that can enter a causal design.

AS THE TREATMENT

Does violent framing reduce support for a protest?

A classifier codes the frame of each news article. The frame is the treatment D.

AS THE OUTCOME

Did a moderation policy reduce hate speech?

A classifier scores every reply before and after the policy. The score is the outcome Y, e.g. in a DiD.

AS A CONTROL

Does coverage change opinion, net of topic?

Embeddings summarise what each article is about, so we compare articles on the same topic. Topic joins X.

The catch

Classifier mistakes become measurement error in the causal estimate, and they are rarely random. Validate the codes against human coding first. That is next week’s topic.

PART 2 · SUMMARY

Four jobs, one rule

Job	The prediction task	Typical tools	What ML cannot do
1 · Adjust for confounders	outcome and treatment from many controls	lasso, random forests, double ML	find confounders we never measured
2 · Sharpen natural experiments	treatment from instruments; the treated unit from donors	lasso, elastic net, matrix completion	make an instrument valid or trends parallel
3 · Find who is affected	the effect for each kind of person	causal trees and forests	explain why the effect differs
4 · Measure from text	codes from documents and images	classifiers, embeddings, LLMs	guarantee that the codes are right

Design first, prediction second.

ML predicts the missing world more flexibly. Whether that world is the right counterfactual depends on the assumptions we argued for in Part 1.

PART 3 · IN THE LAB

From simple comparison to double ML

The question: does being eligible for a company pension plan (401(k)) raise a family's savings? Eligibility is not random: better-paid jobs tend to offer it.

01 Simple comparison: eligible minus not eligible. Selection bias included.

02 Double ML: **ml_l** predicts savings from X, **ml_m** predicts eligibility from X, **n_folds=5** is the cross-fitting.

03 Compare the two numbers. How much of the raw gap was selection?

Install first: pip install doubleml

```
import doubleml as dml
from doubleml.datasets import fetch_401K
from sklearn.ensemble import (RandomForestRegressor,
    RandomForestClassifier)

df = fetch_401K(return_type='DataFrame')
X = ['age', 'inc', 'educ', 'fsize', 'marr',
     'twoearn', 'db', 'pira', 'hown']

# 1. simple comparison
print(df.groupby('e401')['net_tfa'].mean().diff())

# 2. double machine learning
data = dml.DoubleMLData(df, y_col='net_tfa',
    d_cols='e401', x_cols=X)
model = dml.DoubleMLPLR(data,
    ml_l=RandomForestRegressor(n_estimators=500),
    ml_m=RandomForestClassifier(n_estimators=500),
    n_folds=5)
model.fit()
print(model.summary)
```